2 A student investigates the cooling of a liquid in a beaker.

The temperature θ_R of the laboratory is measured using a thermometer.

Hot water is added to an insulated beaker, as shown in Fig. 2.1.

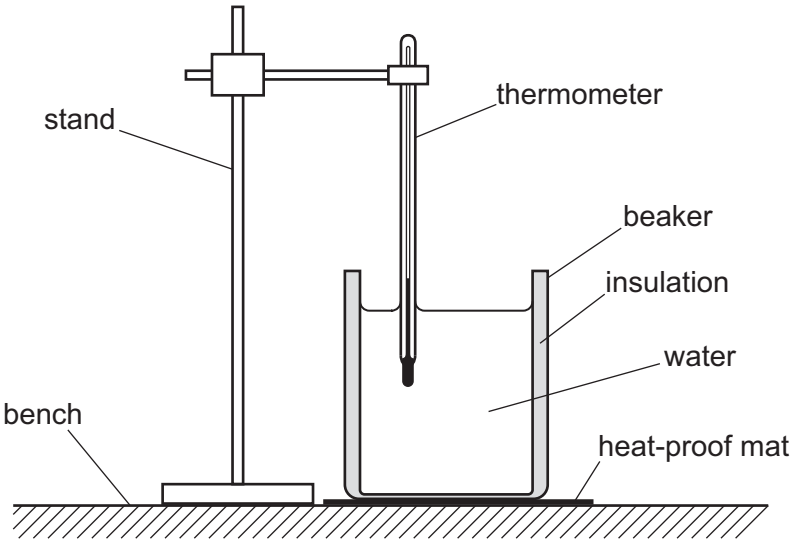


Fig. 2.1

The thermometer measures the temperature of the water. At time t the temperature of the water is θ.

A series of readings of t and θ are taken.

It is suggested that θ and t are related by the equation

$$\theta = \theta_R + (\theta_0 - \theta_R)e^{-(\frac{t}{K})}$$

where θ_0 is the temperature at $t = 0$ and K is a constant.

(a) A graph is plotted of $\ln(\theta - \theta_R)$ on the y-axis against t on the x-axis.

Determine expressions for the gradient and y-intercept.

gradient = ...

y-intercept = ...

[1]

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(b) Values of t and θ are given in Table 2.1.

Table 2.1

t / min	θ / °C	$(\theta - \theta_R)$ / °C	$\ln((\theta - \theta_R)/°C)$
6.0	75.0 ± 0.5		
12.0	64.5 ± 0.5		
18.0	57.0 ± 0.5		
24.0	50.0 ± 0.5		
30.0	44.5 ± 0.5		
36.0	41.0 ± 0.5		

The value of θ_R is (18.5 ± 0.5) °C.

Calculate and record values of $(\theta - \theta_R)$ / °C and $\ln((\theta - \theta_R)/°C)$ in Table 2.1.
Include the absolute uncertainties in $(\theta - \theta_R)$ and $\ln((\theta - \theta_R)/°C)$. [2]

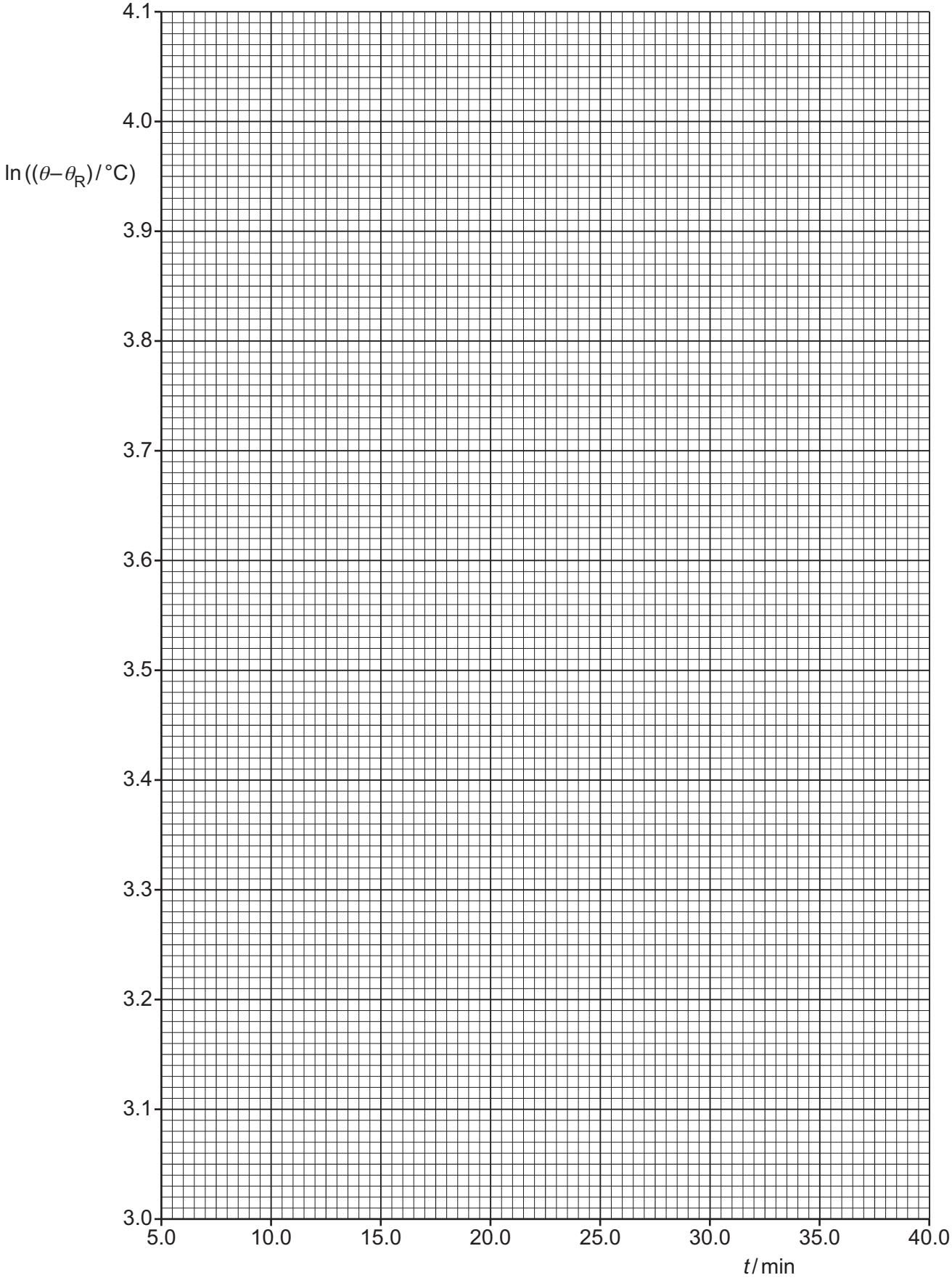
(c) **(i)** Plot a graph of $\ln((\theta - \theta_R)/°C)$ against t / min. Include error bars for $\ln((\theta - \theta_R)/°C)$. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = ... [2]

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(iv) Determine the *y*-intercept of the line of best fit. Include the absolute uncertainty in your answer.

y-intercept = ... [2]

(d) (i) Using your answers to **(a)**, **(c)(iii)** and **(c)(iv)**, determine the values of K and θ_0. Include appropriate units.

K = ...

θ_0 = ...
[2]

(ii) Determine the absolute uncertainty in your value of θ_0.

absolute uncertainty = ... [1]

(e) Determine the time t for the temperature to reach 25.0 °C.

t = ... min [1]

[Total: 15]

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